EE1-10A MATHEMATICS I

1. (a) Express in the form $x+iy$: [Seen similar.]

$$(i)\quad \frac{\sin\frac{\pi}{3}+i\cos\frac{\pi}{3}}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}}, \qquad (ii)\quad \left(\frac{i+1}{i-1}\right)^{2019} \qquad (iii)\quad (-i)^{-i}$$

SOLUTION

Each time, express in complex exponential form:

(i) $$\frac{\sin\frac{\pi}{3}+i\cos\frac{\pi}{3}}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}} = \frac{i(\cos\frac{\pi}{3}-i\sin\frac{\pi}{3})}{\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}} = i\left(\frac{e^{-i\pi/3)}}{e^{i\pi/6}}\right) = ie^{-i\pi/2} = i(-i) = 1$$

(ii) $$\left(\frac{i+1}{i-1}\right)^{2019} = \left(\frac{\sqrt{2}e^{i\pi/4}}{\sqrt{2}e^{i3\pi/4}}\right)^{2019} = \left(e^{-i\pi/2}\right)^{2019} = (-i)^{2019} = (-i)^3(-i)^{2016} = -i$$

(iii) $$(-i)^{-i} = \left(e^{i(-\pi/2+2n\pi)}\right)^{-i} = e^{-i^2(-\pi/2+2n\pi)} = e^{-\pi/2+2n\pi}$$, for integer n.

For part (i), most of the errors were due to a mistake in converting from trigonometric to complex exponential form. For part (ii), most of the errors were due to errors in evaluating the 2019th power. For part (iii), the extra $2n\pi$ was missing.

(b) (i) Use the definitions of sinh and cosh in terms of exponentials, to prove that [Bookwork]

$$\sinh(x+iy) = \sinh x\cos y + i\cosh x\sin y\,.$$

SOLUTION Beginning by expressing as exponentials:

$$\sinh(x+iy) = \frac{e^{x+iy}-e^{-(x+iy)}}{2} = \frac{e^xe^{iy}-e^{-x}e^{-iy}}{2} = \frac{e^x(\cos y+i\sin y)-e^{-x}(\cos y-i\sin y)}{2}$$

$$= \frac{1}{2}\left(e^x-e^{-x}\right)\cos y + i\frac{1}{2}\left(e^x+e^{-x}\right)\sin y = \sinh x\cos y + i\cosh x\sin y\,,$$

as required.

Most mistakes encountered were due to inability to accurately manipulate complex exponentials.

(ii) Hence, or otherwise, find all complex roots of [Seen similar]

$$\sinh^2(z) = -1\,.$$

SOLUTION Begin with $\sinh^2(z) = -1 \Rightarrow \sinh(z) = \sinh(x+iy) = \pm i$, and use (i) to rewrite as

$$\sinh x\cos y + i\cosh x\sin y = \pm i\,.$$

Equating real and imaginary parts:

$$\sinh x\cos y = 0 \quad (1), \quad \text{and} \quad \cosh x\sin y = \pm 1 \quad (2).$$

Begin with (1):

$$\sinh x\cos y = 0 \Rightarrow \sinh x = 0 \quad \text{or} \quad \cos y = 0$$

so either $x = 0$ or $y = (2n+1)\pi/2$ for integer n. Substitute these into (2):

$$x = 0: \quad \cosh 0\sin y = \pm 1 \Rightarrow \sin y = \pm 1 \Rightarrow y = (2n+1)\pi/2\,,$$

so no new solutions here. Similarly,

$$y = (2n+1)\pi/2: \quad \cosh x\sin\left(\frac{(2n+1)\pi}{2}\right) = \pm 1 \Rightarrow \cosh x = \pm 1\,.$$

As $\cosh x = -1$, has no solution and $\cosh x = 1 \Rightarrow x = 0$, we find no new solutions again, leaving $z = (2n+1)\pi i/2$.

Errors were due to the failure to properly expand $sinh(z) = \frac{exp(z)-exp(-z)}{2} = \frac{exp(x+iy)-exp(-(x+iy))}{2}$ **and then determine the appropriate conditions for real and imaginary parts.**

(c) Find the following limits

(i) $\lim_{x\to 3} \dfrac{\sqrt{x-2}-1}{\sqrt{x^2-5}-2}\,,$ [Seen similar]

SOLUTION

Can use l'Hopital, as of form "0/0":

$$\lim_{x\to 3} \frac{\sqrt{x-2}-1}{\sqrt{x^2-5}-2} = \lim_{x\to 3} \frac{\frac{1}{2\sqrt{x-2}}}{\frac{1}{2\sqrt{x^2-5}}2x} = \lim_{x\to 3} \frac{\sqrt{x^2-5}}{2x\sqrt{x-2}} = \frac{\sqrt{4}}{6\sqrt{1}} = \frac{1}{3}\,.$$

(ii) $\lim_{x\to\infty}\left(\frac{x+1}{x}\right)^x$, [New]

SOLUTION

Let $y=\left(\frac{x+1}{x}\right)^x$, then $\ln y = x\ln\left(1+\frac{1}{x}\right)$. As $x\to\infty$, clearly $\left|\frac{1}{x}\right|<1$ so can use series:

$$\ln y = x\left(\frac{1}{x}-\frac{1}{2x^2}+\frac{1}{3x^3}-\cdots\right). = 1-\frac{1}{2x}+\frac{1}{3x^2}-\cdots$$

Hence

$$\lim_{x\to\infty}\ln y = 1 \Rightarrow \lim_{x\to\infty} y = e\,.$$

(iii) $\lim_{x\to 0}\frac{\ln(1-2x^2)}{e^x-x-1}$ [Seen similar]

SOLUTION

Again, given that as $x\to 0$ we have $|-2x^2|<1$ we can apply Maclaurin series in numerator, as well as in denominator:

$$\lim_{x\to 0}\frac{\ln(1-2x^2)}{e^x-x-1} = \lim_{x\to 0}\frac{(-2x^2)-\frac{1}{2}(-2x^2)^2+\frac{1}{3}(-2x^2)^3+\cdots}{1+x+\frac{1}{2}x^2+\frac{1}{6}x^3+\frac{1}{4!}x^4+\cdots-x-1} = \lim_{x\to 0}\frac{-2x^2-2x^4+\frac{8}{3}x^6+\cdots}{\frac{1}{2}x^2+\frac{1}{6}x^3+\frac{1}{4!}x^4+\cdots}$$

$$= \lim_{x\to 0}\frac{-2-2x^2+\frac{8}{3}x^4+\cdots}{\frac{1}{2}+\frac{1}{6}x+\frac{1}{4!}x^2+\cdots} = -4\,.$$

Most of the errors were the result of failure to go beyond a few steps of applying L'Hospital's rule.

(d)

A function is defined as

$$h(x)=\begin{cases} x^{-3}\tan^3(2x) & \text{for} \quad x\neq 0\,; \\ m & \text{for} \quad x=0\,, \end{cases}$$

What value of m makes $h(x)$ a continuous function? [Seen similar]

SOLUTION

For continuity, we need $\lim_{x\to 0}\frac{\tan^3(2x)}{x^3} = h(0) = m$, so we rewrite as

$\lim_{x\to 0}\left(\frac{\sin(2x)}{x}\right)^3 \frac{1}{\cos^3(2)} = 2^3/1^3 = 8$, where we have assumed $\lim_{x\to 0}\sin(ax)/x = a$. Hence for continuity at $x = 0$, we require $m = 8$.

Many errors involved the use of L'Hospital's rule and it made the entire computation more complicated instead of utilizing the limit $\lim_{x\to 0}\sin(x)/x = 1$.

2 (a) Differentiate to obtain $\dfrac{dy}{dx}$: [Seen similar]

$$(i)\quad y=\frac{\ln x}{xe^x}, \qquad (ii)\quad y=\ln(x+\sqrt{x^2-1}), \qquad (iii)\quad x^2+y^2+e^{xy}=1\,.$$

SOLUTION

(i) Quotient rule: $y'=\dfrac{f'g-fg'}{g^2}=\dfrac{\frac{1}{x}(xe^x)-\ln x(e^x+xe^x)}{(xe^x)^2}=\dfrac{1-(1+x)\ln x}{x^2e^x}$.

(ii) Chain rule:

$$y'=\frac{1}{x+\sqrt{x^2-1}}\left(1+\frac{2x}{2\sqrt{x^2-1}}\right)=\frac{1}{x+\sqrt{x^2-1}}\left(\frac{\sqrt{x^2-1}}{\sqrt{x^2-1}}+\frac{x}{\sqrt{x^2-1}}\right)=\frac{1}{\sqrt{x^2-1}}.$$

(iii) Implicit differentiation:

$$2x+2y\frac{dy}{dx}+e^{xy}\left(y+x\frac{dy}{dx}\right)=0\Rightarrow(2y+xe^{xy})\frac{dy}{dx}=-(2x+ye^{xy})\Rightarrow\frac{dy}{dx}=-\frac{2x+ye^{xy}}{2y+xe^{xy}}.$$

For part (i), error involved using $\frac{f'g+fg'}{g^2}$ instead of the quotient rule. Careless errors for parts (ii) and (iii).

(b) Using the definition of the derivative as a limit (differentiation from first principles), differentiate $y=\dfrac{1}{x}$. [Bookwork]

SOLUTION

Differentiating from first principles:

$$\frac{dy}{dx}=\lim_{h\to0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to0}\frac{\frac{1}{x+h}-\frac{1}{x}}{h}=\lim_{h\to0}\frac{x-(x+h)}{x(x+h)h}=\lim_{h\to0}\frac{-1}{x(x+h)}=-\frac{1}{x^2}.$$

The few errors found involved the evaluation of the the limit $\lim_{h\to0}\frac{f(x+h)-f(h)}{h}$. Should be f(x) and not f(h).

(c) Given the function $y=e^{x^2/3}$, [Seen similar/New]

(i) Show that $\dfrac{dy}{dx}=\dfrac{2}{3}xy$.

SOLUTION Using chain rule:

$$\frac{dy}{dx} = e^{x^2/3}\left(\frac{2x}{3}\right) = \frac{2}{3}xy\,.$$

as required.

No problems here.

(ii) Differentiate the last equation n times, using Leibnitz' formula, and obtain a recurrence relation giving $y^{(n+1)}(0)$ in terms of $y^{(n-1)}(0)$.

[Note: $y^{(k)}(a)$ means y differentiated k times, and evaluated at $x = a$.]

SOLUTION

Use Leibnitz' theorem to differentiate n times:

$$D^n(y') = \frac{2}{3}D^n(xy) \Rightarrow y^{(n+1)}(x) = \frac{2}{3}\left(xy^{(n)}(x) + n(1)y^{(n-1)}(x) + 0 + 0 + \ldots\right)$$

Now substitute $x = 0$ to obtain the required recurrence relation:

$$y^{(n+1)}(0) = \frac{2}{3}ny^{(n-1)}(0)\,.$$

Failure to either use Leibnitz's formula (tried to notice a pattern) or to use the Leibnitz formula properly accouned for the majority of errors.

(iii) Hence, or otherwise, evaluate $y^{(5)}(0)$.

SOLUTION To find the fifth derivative $y^{(5)}(0)$, we first need

$$y^{(0)}(0) = y(0) = e^0 = 1 \quad \text{and} \quad y^{(1)}(0) = y'(0) = \frac{2}{3}(0)y = 0\,.$$

Now use the recurrence relation, first with $n = 4$, then $n = 2$:

$$y^{(5)}(0) = \frac{2}{3}(4)y^{(3)}(0) = \frac{8}{3}\frac{2}{3}(2)y^{(1)}(0) = \frac{32}{9}0 = 0\,.$$

Students who got part (ii) wrong are not expected to get full marks for part (iii) as the principle of evaluating the 5th derivative at x = 0 was not achieved.

(d) Prove by induction that for $n \geq 0$,

$$D^n\left[\sin(ax)\right] = a^n \sin\left(ax + \frac{n\pi}{2}\right)\,,$$

where D is the differential operator, $D = \dfrac{d}{dx}$. [Bookwork]

SOLUTION

Let the statement be $P(n)$. Begin with the base case, $n = 0$. The statement is

$$D^0[\sin(ax)] = a^0 \sin\left(ax + \frac{0\pi}{2}\right).$$

the lhs is $\sin(ax)$ differentiated zero times, so $\sin(ax)$, equal to the rhs, establishing the base case, $P(0)$.

For the inductive step, assume the statement is true for $n = k$, the statement

$$P(k): \quad D^k[\sin(ax)] = a^k \sin\left(ax + \frac{k\pi}{2}\right).$$

Then differentiate both sides, so

$$D^{k+1}[\sin(ax)] = \frac{d}{dx} D^k[\sin(ax)] = a^k \frac{d}{dx}\left[\sin\left(ax + \frac{k\pi}{2}\right)\right] = a^k \cdot a\cos\left(ax + \frac{k\pi}{2}\right)$$

and as $\cos\theta = \sin(\theta + \pi/2)$, in general, we have

$$D^{k+1}[\sin(ax)] = a^{k+1} \sin\left(ax + \frac{k\pi}{2} + \frac{\pi}{2}\right) = a^{k+1} \sin\left(ax + \frac{(k+1)\pi}{2}\right)$$

This is the statement for $n = k+1$. Hence, we have shown that $P(k) \Rightarrow P(k+1)$ and, together with the base case $P(0)$, the result is true for all integers $n \geq 0$, by mathematical induction.

Errors mainly resulted from not using the method of induction.

3. (a) Evaluate the following integrals: [Seen similar]

(i) $\displaystyle\int \frac{\sin x-1}{\sin x+1}\,dx,$

SOLUTION

Multiply numerator and denominator by $\sin x-1$:

$$\int \frac{(\sin x-1)(\sin x-1)}{(\sin x+1)(\sin x-1)}\,dx = \int \frac{(\sin^2 x-2\sin x+1}{-\cos^2 x}\,dx = \int -(\tan^x+\sec^2 x)+\frac{2\sin x}{\cos^2 x}\,dx$$

$$\int -(2\sec^x-1)+\frac{2\sin x}{\cos^2 x}\,dx = -2\tan x+x+\frac{2}{\cos x}+C$$

where we have used the identity $\tan^2 x = \sec^2 x-1$. Alternatively, could use substitution $t = \tan(x/2)$.

A lot of people used $t = \tan(x/2)$ substitution and ended up with a difficult integral. Some rewrote the integral as $\int 1-\frac{2}{\sin x+1}\,dx$ and then applied the substitution, which worked.

(ii) $\displaystyle\int \frac{x^2+2}{x^3+x^2-2x}\,dx,$

SOLUTION

Partial fractions first:

$$\frac{x^2+2}{x(x^2+x-2)} = \frac{A}{x}+\frac{B}{x+2}+\frac{C}{x-1} \Rightarrow x^2+2 = A(x-1)(x+2)+Bx(x-1)+Cx(x+2)$$

and letting $x = 0 \Rightarrow 2 = A(-1) \Rightarrow A = -1\,, \quad x = -2 \Rightarrow 4+2 = B(-2)(-3) \Rightarrow B = 1$

and $x = 1 \Rightarrow 1+2 = C(1)(3) \Rightarrow C = 1$, so

$$\int \frac{x+2}{x^3+x^2-2x}\,dx = \int -\frac{1}{x}+\frac{1}{x+2}+\frac{1}{x-1}\,dx = -\ln x+\ln(x+2)+\ln(x-1)+C.$$

A few people tried manipulations to get $\int \frac{f'}{f}\,dx$ but this didn't help much. Only correct solutions with partial fractions. Some arithmetic error in partial fractions

(iii) $\displaystyle\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx,$

SOLUTION

To solve the improper integral, replace the upper limit by k and after integrating, take limit $k \to 1$. Hence let $u = 1 - x^2 \Rightarrow du = -2x\,dx$ and

$$\int \frac{x}{\sqrt{1-x^2}}\,dx = -\frac{1}{2}\int u^{-1/2}\,du = -\frac{1}{2}\frac{u^{1/2}}{1/2} = -\sqrt{u} = -\sqrt{1-x^2}\,.$$

Now apply limits of integration:

$$\int_0^k \frac{x}{\sqrt{1-x^2}}\,dx = -\left[\sqrt{1-x^2}\right]_0^k = -\sqrt{1-k^2} + \sqrt{1}$$

and finally take limit:

$$\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx = \lim_{k\to 1}\int_0^k \frac{x}{\sqrt{1-x^2}}\,dx = \lim_{k\to 1}\left(-\sqrt{1-k^2}+1\right) = 1\,.$$

Very few people saw the improper integral, ignored and lost mark. Many used the substitution $x = \sin u$ which made a proper integral, and got full marks.

(b) Integrate to show that [Seen similar/New]

$$\int \sqrt{1-x^2}\,dx = \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x + C\,.$$

SOLUTION

Need a substitution: answer as given suggests using

$x = \sin u \Rightarrow dx = \cos u\,du$ and $1 - x^2 = 1 - \sin^2 u = \cos^2 u \Rightarrow \sqrt{1-x^2} = \cos u$

so substituting into the integral we get

$$I = \int \sqrt{1-x^2}\,dx = \int \cos^u\,du = \frac{1}{2}\int \cos 2u + 1\,du = \frac{1}{4}\sin 2u + \frac{1}{2}u + c = \frac{1}{2}\sin u\cos u + \frac{1}{2}u + c$$

and as $x = \sin u \Rightarrow u = \sin^{-1}x$ we have $I = \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\sin^{-1}x + c$, as required.

Almost all got this

(c) Given the function $f(x) = \sqrt{1+x}$,

(i) Obtain the first four terms of the Maclaurin series of $f(x)$, and the Lagrange form of the remainder term R_3; [Seen similar]

SOLUTION

The Maclaurin series requires the function and four derivatives:

$$f(x) = (1+x)^{1/2} \Rightarrow f(0) = 1\,,$$

$$f'(x) = \frac{1}{2}(1+x)^{-1/2} \Rightarrow f'(0) = \frac{1}{2}\,,$$

$$f''(x) = -\frac{1}{4}(1+x)^{-3/2} \Rightarrow f''(0) = -\frac{1}{4}\,,$$

$$f^{(3)}(x) = \frac{3}{8}(1+x)^{-5/2} \Rightarrow f^{(3)}(0) = \frac{3}{8}\,,$$

$$f^{(4)}(x) = -\frac{15}{16}(1+x)^{-7/2}\,.$$

The series is therefore

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f^{(3)}(0)}{6}x^3 + R_3 = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + R_3$$

with remainder term

$$R_3(x) = -\frac{5}{128(1+c)^{7/2}}x^4\,,$$

where $0 < |c| < |x|$.

Series almost all correct. A lof of people left the remainder term in the form $f^(4)(c)x^4/4!$ which is fine in theory, but needed the actual function for full marks. Lots of errors involving $1/(1+x)^{7/2}$, a few with wrong derivative or power of x.

(ii) If we use the series obtained in (i) to estimate $\sqrt{1.01}$, what is the maximal error incurred? Explain your reasoning. [New]

SOLUTION

To estimate $\sqrt{1.01} = \sqrt{1+0.01}$, we use $x = 0.01$ so $0 < c < 0.01$. Thus $(1+c)^{7/2}$ is smallest when $c = 0$, which gives the largest error, so the maximal error is

$$E_{\max} \;<\; |R_3| = \frac{5}{128}(0.01)^4 = \frac{5}{128} \times 10^{-8}\,.$$

Most people entered $c = 0$ or $c = 0.01$ with no explanation for their choice, which is needed; sometimes with slightly nebulous explanations. Many people left out the idea of absolute value giving max error including a negative sign. Lots of people used $x = 0.1$, many used $x = 1.01$, and various other combinations.

4. (a) Find the radius of convergence of the following series: [Seen similar]

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n+2} x^n$

SOLUTION Again, the Ratio Test:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+3)\ \ln(n+1)}{(n+2)\ln(n)}\right| |x|$$

Taking the limits:

$$\lim_{n\to\infty} \frac{n+3}{n+1} = \lim_{n\to\infty} \frac{1+\frac{3}{n}}{1+\frac{1}{n}} = 1$$

and for the other limit we use l'Hopital, case "∞/∞",

$$\lim_{n\to\infty} \frac{\ln(n+1)}{\ln n} = \lim_{n\to\infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n}{n+1} = 1$$

which leaves

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = |x| < 1$$

for convergence, and the radius of convergence is $R = 1$.

A lot of people got as far as $\left|\frac{(n+3)\ \ln(n+1)}{(n+2)\ln(n)} x\right|$ but then just jumped straight to the correct, or incorrect answer without looking at the actual limits. Many deduced that $\ln(n+1)/\ln(n) = \ln[(n+1)/n] = 0$, often unable to carry on afterwards. Many wrote down the interval instead of the radius of convergence.

(ii) $\displaystyle\sum_{n=1}^{\infty} \frac{x^{3n}}{n!}$

SOLUTION

Ratio Test: $\displaystyle\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{n!\ x^{3(n+1)}}{(n+1)!\ x^{3n}}\right| = \lim_{n\to\infty} \frac{1}{n+1} |x^3| = 0 < 1$,
for all x, so the series converges for any x and has infinite radius of convergence.

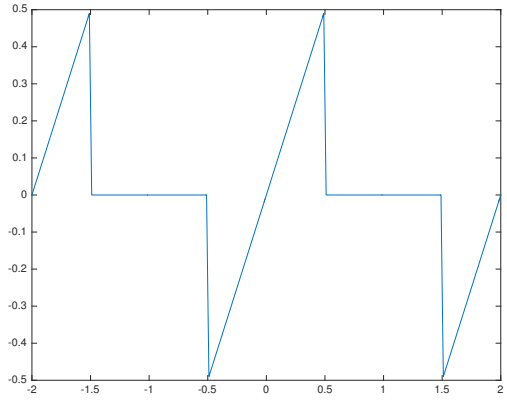
Quite a few people couldn't work out what was meant by the zero in the limit. Some deduced that the function diverged, others just gave up. Many left the radius of convergence in terms of a limit as $n \to \infty$.

(b) Given the function defined on the interval $[0,1)$ as

$$f(t) = \begin{cases} t & 0 \le t < 1/2 \\ 0 & 1/2 \le t < 1 \end{cases}$$

(i) If $g(t)$ is the *odd* extension of $f(t)$, sketch $g(t)$ for $-2 \le t \le 2$. [Seen similar]

SOLUTION



A lof of odd sketches. Most common: putting the first ramp section after the origin, joining points $(1,-1/2)$ to $(2,1/2)$, and similar on negative side. Lots of other variants, many not periodic and/or not odd.

(ii) Show that the first eight terms of the Fourier Series for $g(t)$ are as follows [Seen similar]

$$g(t) = \frac{2}{\pi^2}\sin(\pi t) + \frac{1}{2\pi}\sin(2\pi t) - \frac{2}{(3\pi)^2}\sin(3\pi t) - \frac{1}{4\pi}\sin(4\pi t)$$

$$+\frac{2}{(5\pi)^2}\sin(5\pi t) + \frac{1}{6\pi}\sin(6\pi t) - \frac{2}{(7\pi)^2}\sin(7\pi t) - \frac{1}{8\pi}\sin(8\pi t) + \cdots$$

SOLUTION

The function $g(t)$ is odd, so a_n=0. The period is $T = 2$, so

$$b_n = \int_{-1}^{1} f(t)\ \sin(n\pi t)\ dt\ =\ 2\int_{0}^{1} f(t)\ \sin(n\pi t)\ dt\ =$$

using the half-range series; then

$$b_n = \int_0^{1/2} t\ \sin(n\pi t)\ dt\ +\ 2\int_{1/2}^{1} 0\ \sin(n\pi t)\ dt\ =\ 2\left[-t\frac{\cos(n\pi t)}{n\pi} + \frac{1}{n\pi}\int_0^{1/2}\cos(n\pi t)\ dt\right]$$

$$=\ 2\left[-t\frac{\cos(n\pi t)}{n\pi} + \frac{1}{(n\pi)^2}\sin(n\pi t)\right]_0^{1/2}\ =\ -\frac{1}{n\pi}\cos\left(\frac{n\pi}{2}\right) + 0 + \frac{2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right) - 0$$ In the last expression we note that when n is odd, the cosine terms are zero, while the sine terms give ± 1. Hence

$b_1 = \dfrac{2}{\pi^2}$, $b_3 = -\dfrac{2}{3^2\pi^2}$, $b_5 = \dfrac{2}{5^2\pi^2}$,and $b_7 = -\dfrac{2}{7^2\pi^2}$.

Similarly, for even n the sine terms are zero, and the cosine terms give ± 1 so that

$b_2 = \dfrac{1}{2\pi}$, $b_4 = -\dfrac{1}{4\pi}$, $b_6 = \dfrac{1}{6\pi}$, and $b_8 = -\dfrac{1}{8\pi}$.

The Fourier series is

$$g(t) = \sum_{n=1}^{\infty} b_n \sin(n\pi t)$$

so the first eight terms are

$$g(t) = \frac{2}{\pi^2}\sin(\pi t) + \frac{1}{2\pi}\sin(2\pi t) - \frac{2}{(3\pi)^2}\sin(3\pi t) - \frac{1}{4\pi}\sin(4\pi t) + \frac{2}{(5\pi)^2}\sin(5\pi t) + \frac{1}{6\pi}\sin(6\pi t)$$

$$- \frac{2}{(7\pi)^2}\sin(7\pi t) - \frac{1}{8\pi}\sin(8\pi t) + \cdots$$

as required.

Quite a few people got correctly to $-\frac{1}{n\pi}\cos\left(\frac{n\pi}{2}\right) + \frac{2}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)$ and then couldn't work further, or got some error and then tried usually to make their results fit the solution given in the question. Very common: taking $\left[-t\frac{\cos(n\pi t)}{n\pi}\right]_0^{1/2} = 0$. Also common: trying to fit both sine and cosine expression into a single term, rather than splitting into even and odd cases, usually incurring some error here, though a few did achieve a complex but correct expression!

(iii) Choosing an appropriate value for t in the Fourier series for $g(t)$ above, obtain the value of the infinite sum [Seen similar/New]

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} \cdots$$

SOLUTION

To obtain the desired squares of odd integers in the denominators above, we clearly need only the odd b_n in the series, with even $b_n = 0$. Need to choose t so that

1) sine terms give correct $\pm$ pattern for odd n and

2) sine terms vanish for even $n = 2m$.

The last requirement suggests using $\sin(n\pi) = 0$, so in $\sin(2m\pi t)$, using $t = \dfrac{1}{2}$ works. In the series we now have

$$g(1/2) = \frac{2}{\pi^2}\sin(\pi/2) + 0 - \frac{2}{(3\pi)^2}\sin(3\pi/2) - 0 + \frac{2}{(5\pi)^2}\sin(5\pi/2) + 0 - \frac{2}{(7\pi)^2}\sin(7\pi/2) - 0 + \cdots$$

and we already used the idea of $\sin(n\pi/2) = \pm 1$ for odd n above, leaving

$$g(1/2) = \frac{2}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \cdots\right)$$

Finally, we recognize that $g(1/2)$ is at the discontinuity, where the Fourier series converges to the average of the two values 0 and $1/2$, hence

$$g(1/2) = \frac{1}{4} = \frac{2}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} + \cdots\right)$$

and the result for the series is $\pi^2/8$.

Lots of correct solutions with little or no reasoning given. Many solutions correct on rhs, but ignoring discontinuity on lhs, giving $g(1/2) = 1/2$, or zero.